UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
2020/2021 FIRST SEMESTER EXAMINATION

Session First Semester 2025-2026 Session
COURSE: MTH413: FUNCTIONAL ANALYSIS
TIME: TWO HOURS (2HRS)

Instructions: Attempt any **Three** (3) questions only.

1. (a) Define the following with examples: (i) field and linear space; (ii) scalar field (iii) complete normed linear space; (iii) Norm function (iv) sublinear functionals.

2. (a) Prove that every normed linear space is is a metric space. and define the following terms in normed linear spaces: (i) Limit of sequence; (ii) convergent sequence; (iii) Cauchy sequence;
(b) Prove that if $\{x_1, ..., x_n, ...\}$ is a linearly independent set of vectors in a normed space $(X, \|.\|)$; then there exists a number $c > 0$ such that for every choice of scalars $\alpha_1, \alpha_2, \cdots, \alpha_n$ an we have

$$\|\alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n\| \geq c(|\alpha_1| + |\alpha_2| + \cdots + |\alpha_n|)$$

3. (a) Use the parallelogram law to show that the space $C[0,1]$ with the norm $\|.\|_\infty = \sup_{t\in[0,1]} |.|$ is not an inner product space.
(b) Prove that every convergence sequence is a Cauchy sequence and use this to discourse the concept of Banach and Hilbert spaces.

4. Use any suitable lemma that you know to prove that every finite dimensional subspace Y of a normed space $(X, \|.\|)$ is complete;

5. From the last equation to verify, particularly, that every finite dimensional normed space is a Banach space.